

# Erdős Problem #983

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| EP-983 | L1 - Tractable | Partiellement résolu | Number Theory |
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## Énoncé original (anglais, reproduction mot pour mot)

Let  $n \geq 2$  and  $\pi(n) < k \leq n$ . Let  $f(k, n)$  be the smallest integer  $r$  such that in any  $A \subseteq \{1, \dots, n\}$  of size  $|A| = k$  there exist primes  $p_1, \dots, p_r$  such that  $> r$  many  $a \in A$  are only divisible by primes from  $\{p_1, \dots, p_r\}$ .

Is it true that  $2\pi(n^{1/2}) - f(\pi(n) + 1, n) \rightarrow \infty$  as  $n \rightarrow \infty$ ?

In general, estimate  $f(k, n)$ , particularly when  $\pi(n) + 1 < k = o(n)$ .

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